

PRAKTIKUM SISTEM OPERASI

MODUL 15

Keamanan Linux



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

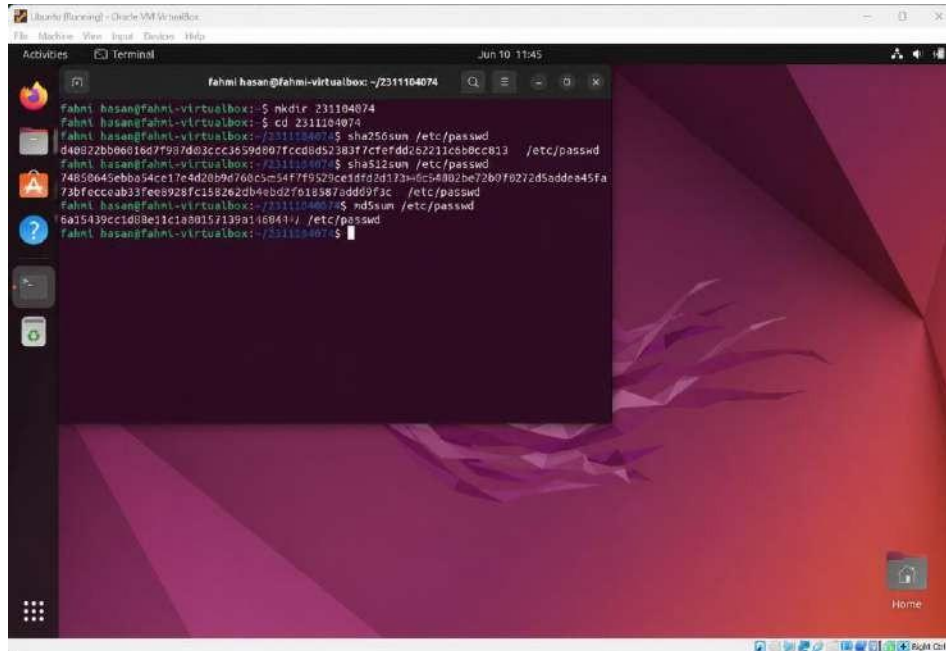
UNIVERSITAS TELKOM

2026

JURNAL

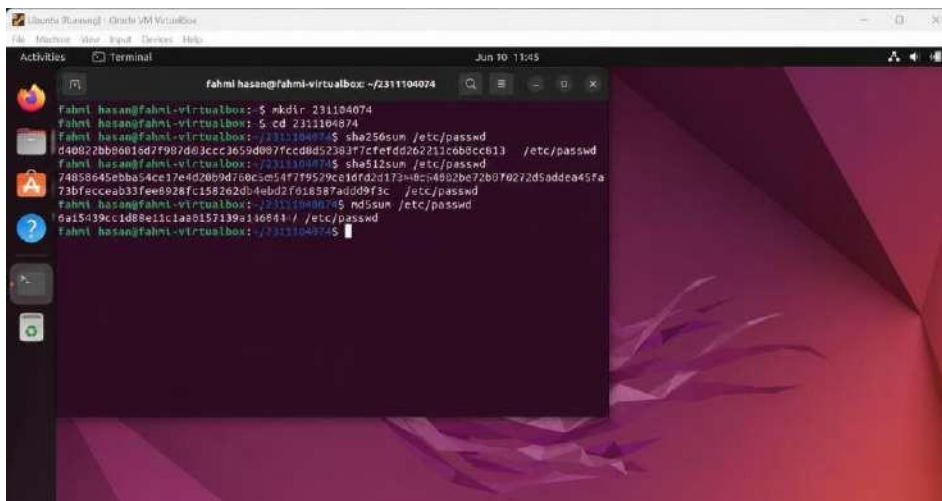
1. Integritas: Dasar Hashing

a. hash SHA256, SHA512, dan MD5



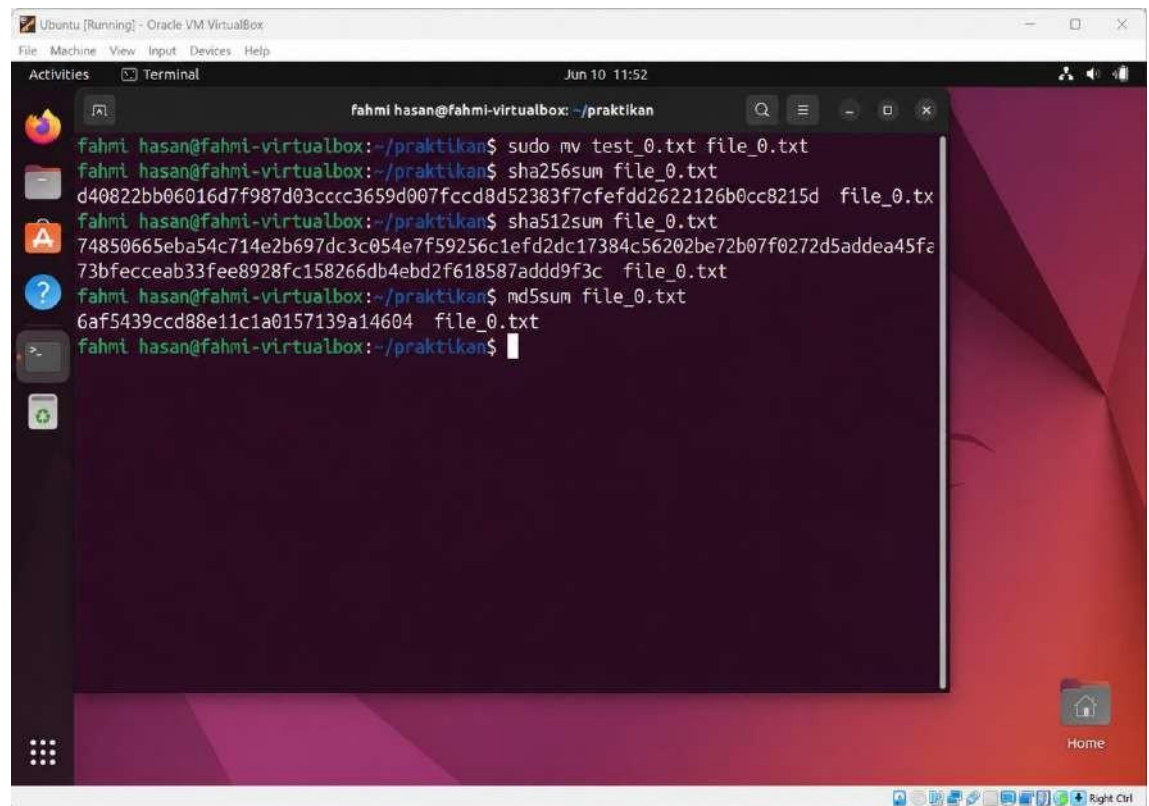
```
fahmi hasan@fahmi-virtualbox: ~/2311104074
fahmi hasan@fahmi-virtualbox: $ mkdir 2311104074
fahmi hasan@fahmi-virtualbox: $ cd 2311104074
fahmi hasan@fahmi-virtualbox: ~/2311104074 $ sha256sum /etc/passwd
d40822bb6016d7f987d03ccc3659d007fccc0d5238377cferdd262211c0b0cc613  /etc/passwd
fahmi hasan@fahmi-virtualbox: ~/2311104074 $ sha512sum /etc/passwd
74850645ebba54ce17e4d20b9d700c5m54f7f9529ce1d02d173m6c54002be72b070272d5adde45fa
73bf0ccea33fee028fc158262db4ebd2f018387add09f3c  /etc/passwd
fahmi hasan@fahmi-virtualbox: ~/2311104074 $ md5sum /etc/passwd
6a15439cc1d88e1c1a0157139a140644 /etc/passwd
fahmi hasan@fahmi-virtualbox: ~/2311104074 $
```

b. File test_0.txt



```
fahmi hasan@fahmi-virtualbox: ~/2311104074
fahmi hasan@fahmi-virtualbox: $ mkdir 2311104074
fahmi hasan@fahmi-virtualbox: $ cd 2311104074
fahmi hasan@fahmi-virtualbox: ~/2311104074 $ sha256sum /etc/passwd
d40822bb6016d7f987d03ccc3659d007fccc0d5238377cferdd262211c0b0cc613  /etc/passwd
fahmi hasan@fahmi-virtualbox: ~/2311104074 $ sha512sum /etc/passwd
74850645ebba54ce17e4d20b9d700c5m54f7f9529ce1d02d173m6c54002be72b070272d5adde45fa
73bf0ccea33fee028fc158262db4ebd2f018387add09f3c  /etc/passwd
fahmi hasan@fahmi-virtualbox: ~/2311104074 $ md5sum /etc/passwd
6a15439cc1d88e1c1a0157139a140644 /etc/passwd
fahmi hasan@fahmi-virtualbox: ~/2311104074 $
```

c. Rename file



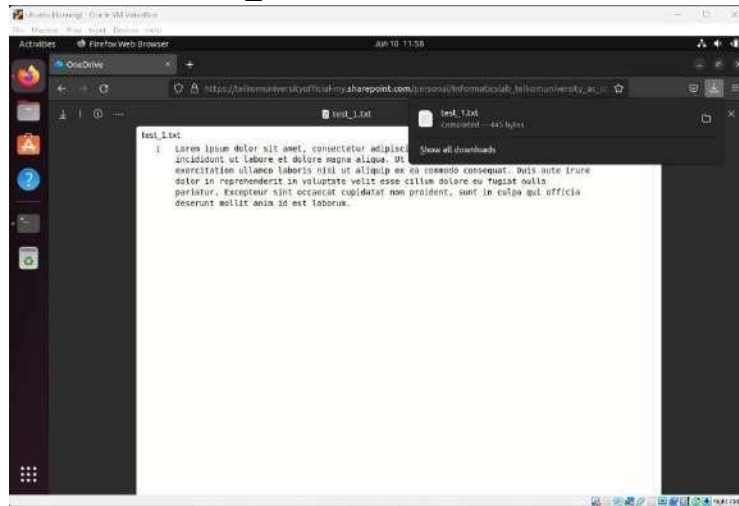
The screenshot shows a terminal window titled 'fahmi hasan@fahmi-virtualbox: ~/praktikan'. The terminal output is as follows:

```
fahmi hasan@fahmi-virtualbox:~/praktikan$ sudo mv test_0.txt file_0.txt
fahmi hasan@fahmi-virtualbox:~/praktikan$ sha256sum file_0.txt
d40822bb06016d7f987d03cccc3659d007fccd8d52383f7cfefdd2622126b0cc8215d  file_0.tx
fahmi hasan@fahmi-virtualbox:~/praktikan$ sha512sum file_0.txt
74850665eba54c714e2b697dc3c054e7f59256c1efd2dc17384c56202be72b07f0272d5addea45fa
73bfecceab33fee8928fc158266db4ebd2f618587add9f3c  file_0.txt
fahmi hasan@fahmi-virtualbox:~/praktikan$ md5sum file_0.txt
6af5439ccd88e11c1a0157139a14604  file_0.txt
fahmi hasan@fahmi-virtualbox:~/praktikan$
```

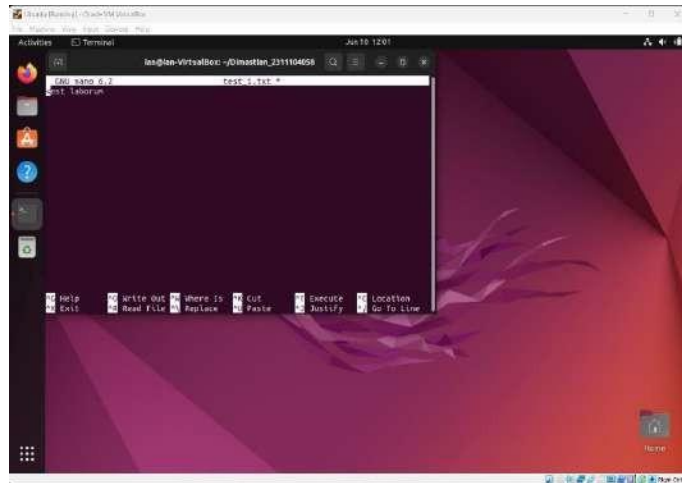
- d. File `/etc/passwd`, `test_0.txt`, dan `file_0.txt` akan memiliki nilai hash yang persis sama. Penjelasannya, algoritma hash menghitung nilainya murni berdasarkan isi (konten/data) file, bukan berdasarkan nama atau lokasi direktorinya. Selama satu bit pun isi datanya tidak berubah, nilai integritas (hash) akan selalu identik.

2. Integritas: Avalance

a. Download file test_1.txt



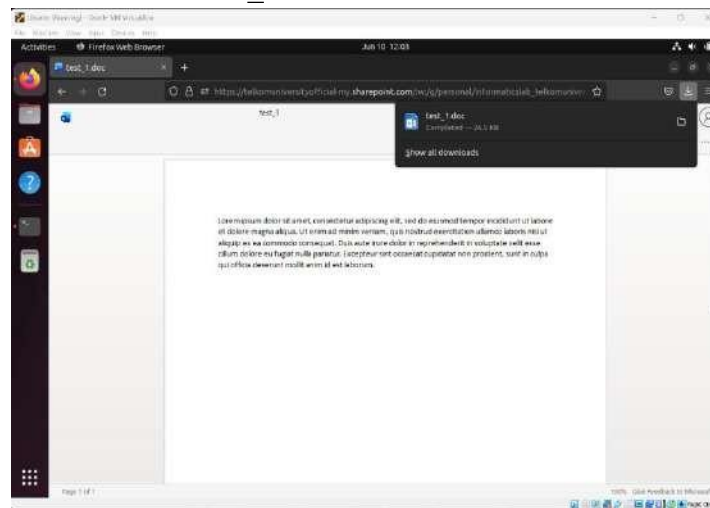
b. Hapus titik di akhir



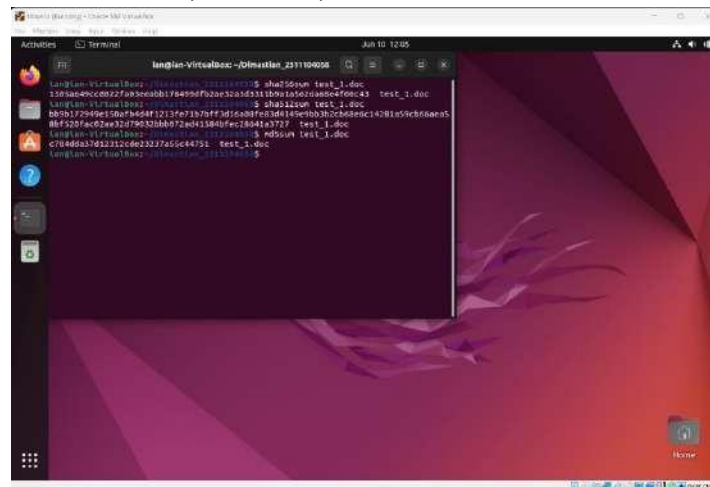
- c. Nilai hash akan berubah total atau sangat berbeda. Meskipun modifikasinya sangat kecil (hanya menghilangkan sebuah tanda titik), algoritma kriptografi bekerja melalui apa yang disebut Avalanche Effect. Setiap perubahan satu karakter/bit sekecil apa pun akan menghasilkan keluaran (hash) yang benar-benar baru. Hash yang tidak cocok menandakan bahwa integritas asal data sudah berubah/terkompromi.

3. Integritas: Metadata

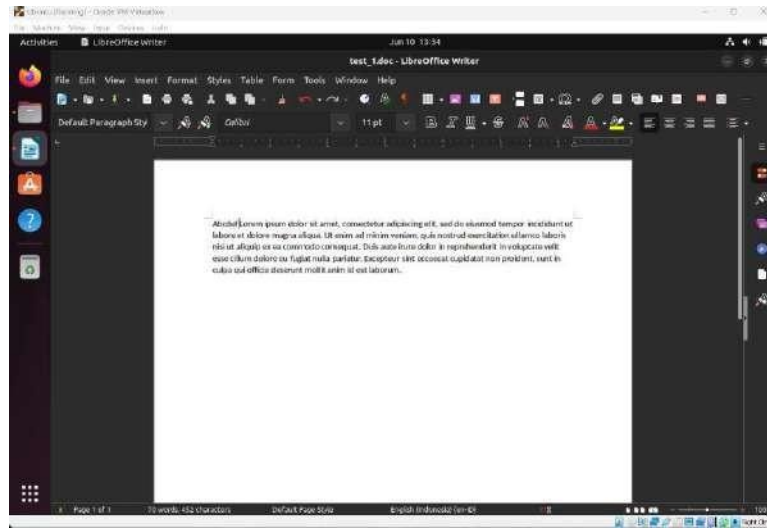
a. Download file test_1.doc



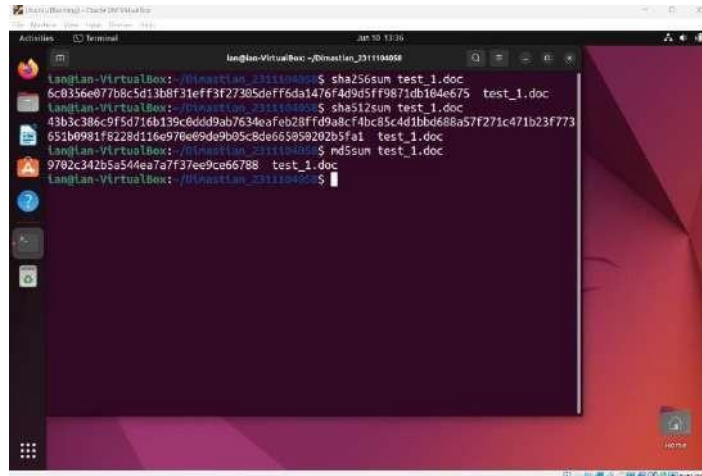
b. hash SHA256, SHA512, dan MD5



c. Ketik huruf



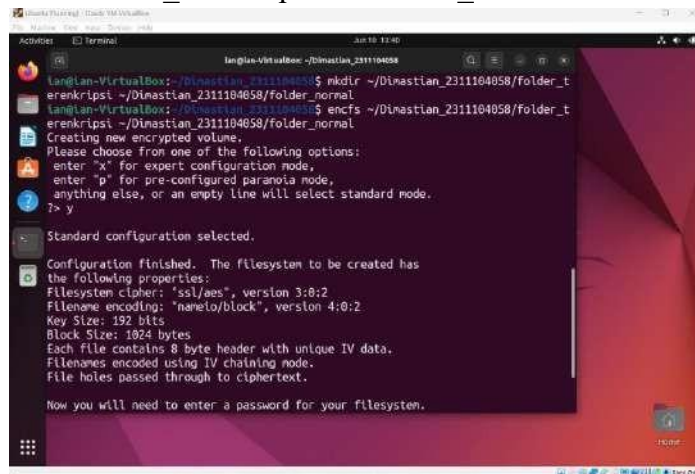
d. hash SHA256, SHA512, dan MD5



- e. Nilai hash yang baru akan menjadi berbeda dari aslinya, walaupun teks di dalamnya sekilas sudah dikembalikan seperti awal. Hal ini terjadi karena format file seperti .doc menyimpan lebih dari sekadar tulisan, ia juga menyimpan data latar belakang berupa Metadata (misal: atribut waktu file dimodifikasi, siapa penulisnya, dll). Setiap kali tombol Save ditekan, atribut "waktu modifikasi terakhir" (last modified timestamp) akan terbaru. Perubahan waktu save ini akan mengubah deretan bit dasar file yang memicu perubahan pada nilai hash yang dihasilkan.

4. Konfidensialitas: encfs

a. encfs folder_terenkripsi dan folder_normal



```
langlan-VirtualBox: ~/Dinastian_2311104058
langlan-VirtualBox:~/Dinastian_2311104058$ mkdir ~/Dinastian_2311104058/folder_tere
langlan-VirtualBox:~/Dinastian_2311104058$ encfs ~/Dinastian_2311104058/folder_tere
langlan-VirtualBox:~/Dinastian_2311104058$ encfs ~/Dinastian_2311104058/folder_tere
Creating new encrypted volume.
Please choose from one of the following options:
enter "x" for expert configuration mode,
enter "p" for pre-configured paranoia mode,
anything else, or an empty line will select standard mode.
?> y

Standard configuration selected.

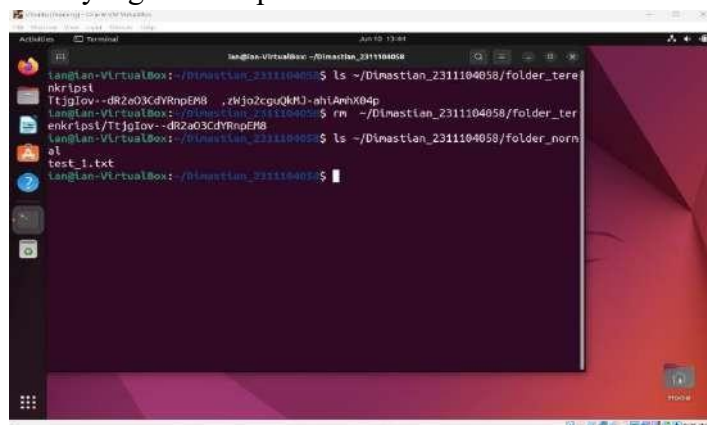
Configuration finished. The filesystem to be created has
the following properties:
Filesystem cipher: 'ssl/aes', version 3:0:2
Filename encoding: 'nameio/block', version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
Filenames encoded using IV chaining mode.
File holes passed through to ciphertext.

Now you will need to enter a password for your filesystem.
```

b. File asli otomatis memiliki nama yang diacak/terenkripsi, dan apabila filenya coba dibaca akan berupa sekumpulan aksara acak (ciphertext). Hal ini merupakan bentuk perlindungan konfidensialitas.

c. Hapus file

File aslinya di folder_normal juga ikut hilang/terhapus. Karena folder_normal pada hakikatnya tidak benar-benar menyimpan file, melainkan hanya bertindak sebagai jendela tempat file yang tidak terenkripsi ditampilkan, sementara wujud asli yang tersimpan secara sistem terletak di folder terenkripsinya.



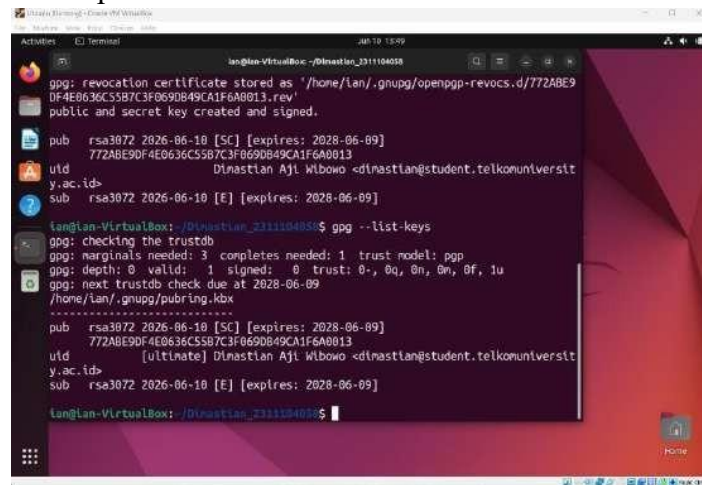
```
langlan-VirtualBox: ~/Dinastian_2311104058
langlan-VirtualBox:~/Dinastian_2311104058$ ls ~/Dinastian_2311104058/folder_teredenkripsi
TtjgIov--dR2a03CdVRnpEN8 .zhjoZcgugkMJ-ahAmhK04p
langlan-VirtualBox:~/Dinastian_2311104058$ rm ~/Dinastian_2311104058/folder_teredenkripsi
langlan-VirtualBox:~/Dinastian_2311104058$ ls ~/Dinastian_2311104058/folder_normal
test_1.txt
langlan-VirtualBox:~/Dinastian_2311104058$
```

d. Unmount

folder_normal seketika berubah menjadi kosong. Seluruh file secara aman dikunci dalam bentuk tak terbaca di dalam folder_terenkripsi. File normal tersebut baru bisa dilihat dan diakses kembali setelah kamu melakukan perintah

5. Konfidensialitas: gpg

a. Kunci publik



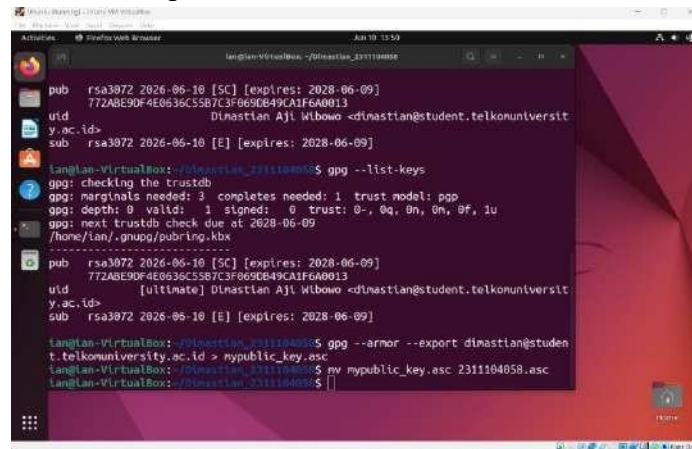
```
lan@lan-VirtualBox: ~/Dinastian_2311104058$ gpg --revocation-certificate stored as '/home/lan/.gnupg/openpgp-revocs.d/772ABE9DF4E0636C55B7C3F0690B49CA1F6A0013.rev'
public and secret key created and signed.

pub   rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
       772ABE9DF4E0636C55B7C3F0690B49CA1F6A0013
uid     [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniverst
y.ac.id>
sub   rsa3072 2026-06-10 [E] [expires: 2028-06-09]

lan@lan-VirtualBox: ~/Dinastian_2311104058$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0f, 1u
gpg: next trustdb check due at 2028-06-09
/home/lan/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
       772ABE9DF4E0636C55B7C3F0690B49CA1F6A0013
uid     [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniverst
y.ac.id>
sub   rsa3072 2026-06-10 [E] [expires: 2028-06-09]

lan@lan-VirtualBox: ~/Dinastian_2311104058$
```

b. Hasil kunci publik



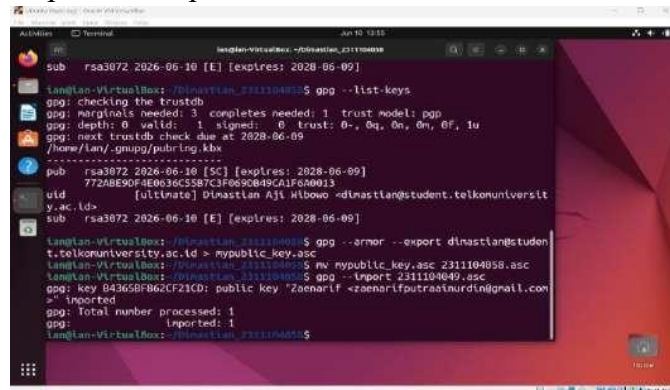
```
lan@lan-VirtualBox: ~/Dinastian_2311104058$ gpg --list-keys
pub   rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
      772ABE90F4E0636C55B7C3F0690B49CA1F6A0013
uid   [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniversit
y.ac.id>
sub   rsa3072 2026-06-10 [E] [expires: 2028-06-09]

lan@lan-VirtualBox: ~/Dinastian_2311104058$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-06-09
/home/lan/.gnupg/pubring.kbx
-----
pub   rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
      772ABE90F4E0636C55B7C3F0690B49CA1F6A0013
uid   [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniversit
y.ac.id>
sub   rsa3072 2026-06-10 [E] [expires: 2028-06-09]

lan@lan-VirtualBox: ~/Dinastian_2311104058$ gpg --armor --export dinastian@studen
t.telkomuniversity.ac.id > mypublic_key.asc
lan@lan-VirtualBox: ~/Dinastian_2311104058$ mv mypublic_key.asc 2311104058.asc
lan@lan-VirtualBox: ~/Dinastian_2311104058$
```

c. Export kunci publik

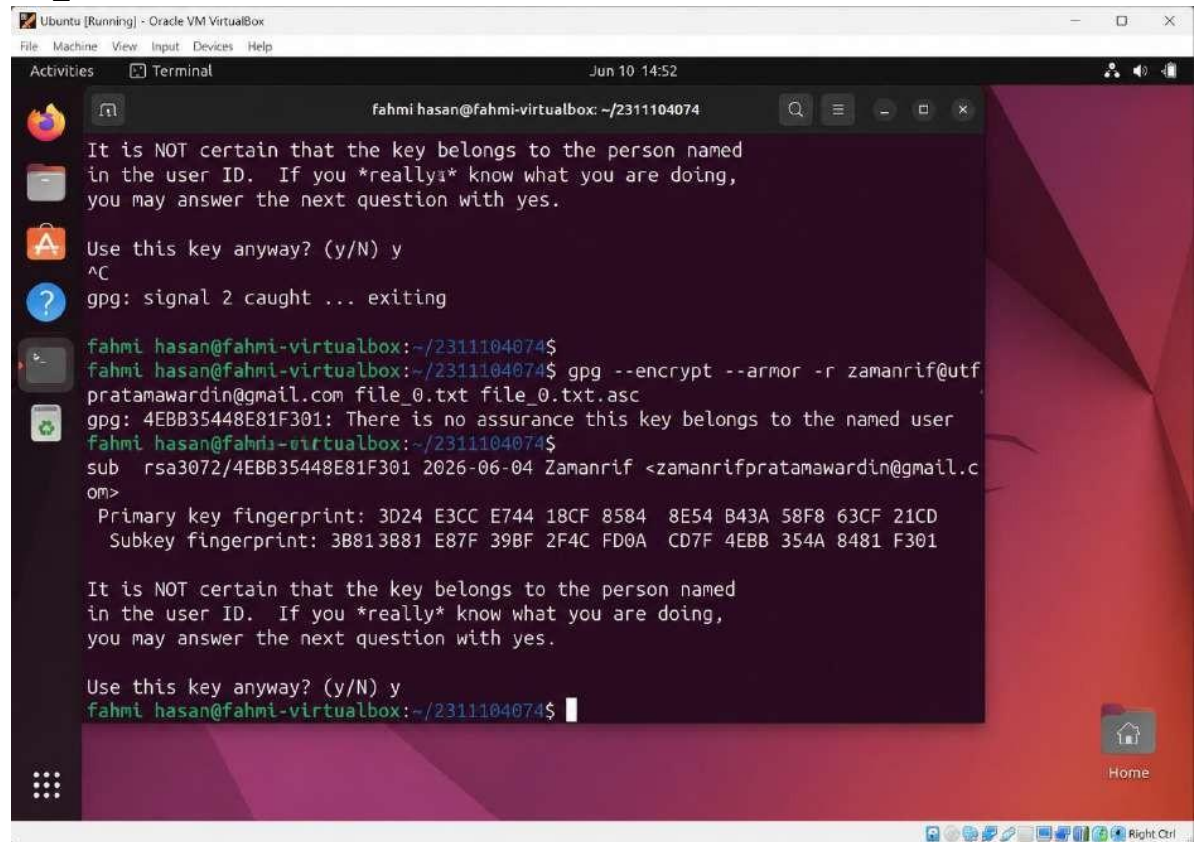
d. Import kunci publik



```
lan@lan-VirtualBox: ~/Dinastian_2311104058$ gpg --list-keys
pub   rsa3072 2026-06-10 [SC] [expires: 2028-06-09]
      772ABE90F4E0636C55B7C3F0690B49CA1F6A0013
uid   [ultimate] Dinastian Aji Wibowo <dinastian@student.telkomuniversit
y.ac.id>
sub   rsa3072 2026-06-10 [E] [expires: 2028-06-09]

lan@lan-VirtualBox: ~/Dinastian_2311104058$ gpg --armor --export dinastian@studen
t.telkomuniversity.ac.id > mypublic_key.asc
lan@lan-VirtualBox: ~/Dinastian_2311104058$ mv mypublic_key.asc 2311104058.asc
lan@lan-VirtualBox: ~/Dinastian_2311104058$ gpg --import 2311104058.asc
gpg: key B4365BFB62CF21CD: public key "Zaenarif <zaenarifoutraaaurding@gmail.com
>" imported
gpg: Total number processed: 1
gpg:   imported: 1
lan@lan-VirtualBox: ~/Dinastian_2311104058$
```

e. file_rahasia



The screenshot shows a terminal window titled "fahmi hasan@fahmi-virtualbox: ~/2311104074" running on an Ubuntu desktop. The terminal displays the following text:

```
It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
^C
gpg: signal 2 caught ... exiting

fahmi hasan@fahmi-virtualbox:~/2311104074$
fahmi hasan@fahmi-virtualbox:~/2311104074$ gpg --encrypt --armor -r zamanrif@utf
pratamawardin@gmail.com file_0.txt file_0.txt.asc
gpg: 4EBB35448E81F301: There is no assurance this key belongs to the named user
fahmi hasan@fahmi-virtualbox:~/2311104074$
sub rsa3072/4EBB35448E81F301 2026-06-04 Zamanrif <zamanrifpratamawardin@gmail.c
om>
Primary key fingerprint: 3D24 E3CC E744 18CF 8584 8E54 B43A 58F8 63CF 21CD
Subkey fingerprint: 3B813881 E87F 39BF 2F4C FD0A CD7F 4EBB 354A 8481 F301

It is NOT certain that the key belongs to the person named
in the user ID. If you *really* know what you are doing,
you may answer the next question with yes.

Use this key anyway? (y/N) y
fahmi hasan@fahmi-virtualbox:~/2311104074$
```

The terminal window is part of an Oracle VM VirtualBox environment, as indicated by the title bar. The desktop background is a red and purple geometric pattern. The system clock shows "Jun 10 14:52".